



Project Proposal

“CareerFlow AI”

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CareerFlow AI - Project Proposal

1. Introduction

CareerFlow AI is an AI-powered job recruitment platform designed to streamline the hiring process for both job seekers and hiring managers. The platform leverages a hybrid AI approach combining Google Gemini and Groq AI to automate CV screening, job matching, interview scheduling, and meeting summaries. It eliminates manual inefficiencies in recruitment by providing an end-to-end automated hiring solution.

2. Problem Statement

Stakeholder	Challenges
Job Seekers	Waste hours sifting through irrelevant job postings; manual CV submissions to multiple companies; no structured feedback
Hiring Managers	Manually screening hundreds of CVs; spending hours shortlisting candidates; back-and-forth emails for interview scheduling
Professionals	Lose 10-15 minutes per meeting on manual scheduling, note-taking, and follow-up emails

2.1 Negative Outcomes

1. Low productivity
2. Burnout
3. Slower career growth
4. Delayed hiring decisions

3. Proposed Solution

CareerFlow AI provides an all-in-one platform that automates the entire recruitment lifecycle:

3.1 For Job Seekers

1. One-time CV upload + preferred job categories
2. Auto job matching & one-click apply
3. CV automatically sent to hiring manager
4. Auto Google Calendar + Meet scheduling
5. Voice recording → Whisper + Groq AI summary
6. Auto follow-up emails + dashboard

3.2 For Hiring Managers

1. Post jobs with Title, Description, Requirements, Salary
2. AI extracts skills + ranks top applicants
3. Auto notification: "This candidate is the top applicant"
4. One-click interview scheduling
5. Smart meeting management with AI transcription

4. Features

4.1 Job Seeker Features

Feature	Description
One-Time Onboarding	Upload CV once and select preferred job categories
Smart Job Matching	Browse only relevant job postings matching your skills
One-Click Apply	Apply instantly — CV automatically submitted
AI-Powered Ranking	Get automatically ranked among all applicants
Automated Scheduling	Receive Google Meet link + Calendar invite directly

4.2 Hiring Manager Features

Feature	Description
Easy Job Posting	Post jobs with Title, Description, Requirements & Salary
Automated Management	Receive all applications in one dashboard with AI-extracted skills
Top Applicant Identification	System automatically highlights Top Applicant(s)
One-Click Scheduling	Select candidate → AI generates Google Meet link + Calendar event
Smart Meeting Management	Record interview → AI transcribes + generates structured summary

4.3 Shared Features

- 1) Auto Google Calendar + Meet scheduling
- 2) Voice recording → Whisper + Groq AI summary
- 3) Auto follow-up emails + dashboard
- 4) Everything organized in one dashboard

5. Hybrid AI Approach

5.1 Gemini API

- 1) Analyzes CV, extracts skills/experience
- 2) Scores job matches (80–100% relevance)
- 3) Avoids duplicate jobs

5.2 Groq AI + Whisper

- 1) Real-time speech-to-text transcription
- 2) Structured JSON summary (key points, decisions, action items with assignees & deadlines)

5.3 Hybrid Advantage

- 1) More accurate and useful than single AI model
- 2) Gemini understands career profiles
- 3) Groq creates smarter meeting summaries tied to job search
- 4) Automated follow-up tasks with Firebase integration

6. Technology Stack

Component	Technology
Frontend	React + TypeScript
Backend	Express.js (Node.js)
Database	Firebase (user data, jobs, meetings, history)
Speech AI	Whisper (speech-to-text)

Component	Technology
AI Models	Google Gemini API + Groq AI (LLaMA 3.3 70B)
Integrations	Google OAuth, Calendar API, Gmail API, Google Meet
Output	Auto PDF/Word download + scheduled emails

7. Database Schema

Collection	Description
Candidates Application	Job applications with status
Candidates Profile	User profiles with CV data
Jobs	Job postings with details
Meetings	Interview schedules and summaries
Users	All platform users

8. Email Notifications

Type	Recipient	Description
Meeting Invitation	Candidate	Meeting details with Google Meet link
Application Received	Hirer	Notification when candidate applies
Meeting Summary	Both	AI-generated summary with action items
Error Notification	Hirer	Incorrect meeting information alert

9. Sample Use Cases

UC-01: Job Seeker Applies for Job

- 1) User uploads CV once
- 2) User selects job categories
- 3) System recommends matching jobs
- 4) User clicks "Apply"

- 5) CV auto-sent to hiring manager
- 6) System tracks application status

UC-02: Hiring Manager Posts Job

- 1) Manager fills job form
- 2) AI extracts required skills
- 3) Job posted to platform
- 4) Applications auto-collected
- 5) AI ranks applicants
- 6) Manager notified of top applicant

UC-03: Interview Scheduling

- 1) Manager selects candidate
- 2) System suggests available time slots
- 3) Calendar invite + Meet link generated
- 4) Emails sent to all attendees
- 5) Event added to all calendars

UC-04: Meeting Summary Generation

- 1) Interview recorded
- 2) Whisper transcribes audio
- 3) Groq AI generates structured summary
- 4) Summary sent to all attendees
- 5) Action items added to dashboard

10. Benefits

Benefit	Description
Time Savings	Saves hours for both job seekers and hiring managers
Higher Success Rate	Increases interview success rate & faster hiring
Centralized Dashboard	Everything organized in one dashboard (no switching apps)
Better Decisions	AI-powered insights for smarter hiring decisions
Automated Follow-ups	No manual email tracking or reminders needed

11. Limitations

Limitation	Description
Platform Dependency	Requires internet connectivity
AI Accuracy	Depends on quality of CV and job descriptions
Google Integration	Requires Google account for Calendar/Gmail
Voice Quality	Transcription accuracy depends on audio quality

12. Future Enhancements

Enhancement	Description
Mobile App	Android/iOS applications
Video Interviews	AI analysis of video interviews
Advanced Analytics	Hiring insights and trends dashboard
Company Branding	Customizable company profiles
Multi-language	Support for Urdu and other regional languages

13. Conclusion

CareerFlow AI revolutionizes the recruitment process by leveraging a hybrid AI approach to automate CV screening, job matching, interview scheduling, and meeting summaries. By combining Gemini's profiling capabilities with Groq's summarization power, the platform provides a seamless, efficient, and intelligent hiring experience for both job seekers and hiring managers.